

Hostname	Message	Date	Source Type	Target Type	Object Class	Permission	User/Context	Command	Detail
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579
localhost	denied {read} for pid=3579	Aug 07 12:44:54	process	file	file	read	httpd_t	httpd -k start	denied {read} for pid=3579

Figure 1: The audit tool

SELinux Log files

If the Linux Auditing System (the `auditd` daemon) is running, SELinux denials are logged into the audit log file. The default audit log file is `/var/log/audit/audit.log`. In a situation where the `auditd` daemon is not running, AVC denials are logged in `/var/log/messages`.

In my opinion, it is prudent to enable the `auditd` daemon and maintain a separate log file. The `audit` RPM should be installed by default on most Red Hat Enterprise systems.

If not installed, do so. In my system running RHEL 5, the RPM is `audit-1.3.1-1.el5`.

Also, do ensure that the `auditd` daemon is enabled to start at boot time. You can do that by issuing the following command:

```
[root@web ~]# chkconfig auditd on
```

To ensure that the `auditd` daemon has been enabled at start-up, issue the following command:

```
[root@web ~]# chkconfig --list auditd
auditd 0:off 1:off 2:on 3:on 4:on 5:
on 6:off
```

As you can see, `auditd` will be executed whenever my system boots in run levels 3 or 5.

Once `auditd` is running, SELinux

logs are written to the audit log file (generally `/var/log/audit/audit.log`).

A typical snippet of an AVC denial message, apart from the time stamp in my log file, is given below:

```
avc: denied {read} for pid=3579
comm="httpd" name="index.html" dev=hd3
ino=33001 scontext=user_u:system_r:httpd_t:s0
tcontext=system_u:object_r:tmp_t:s0 tclass=file
```

Let us look at the log and decipher its meaning:

```
avc: [result of SELinux rules] { access type
} for pid="subject - Process ID Number"
comm="subject - program command"
name="subject - name" dev="object - device"
ino="object - inode number" scontext="subject
- Security Context of Source" tcontext="object
- Security Context of Target" tclass="object
- Class Type of target"
```

The above error can be interpreted by classifying the subject, object details and understanding the operation that the subject was performing on the object.

- Subject details in the log file:
 - Subject Command = `httpd`
 - Subject PID = `3579`
 - Subject Security Context = `user_u:system_r:httpd_t:s0`
- Object details in the log file:
 - Object Name = `index.html`
 - Object Class = `file`
 - Object Partition = `hda3`

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